VISHWAKARMA UNIVERSITY
Maximising Human Potential

VISHWAKARMA UNIVERSITY, PUNE • DEPARTMENT OF COMPUTER ENGINEERING

Cloud-Based Hospital Appointment System

An Enterprise Multi-Tier Healthcare SaaS with Relational Scheduling, Real-Time Doctor Allocations & AWS Cloud Containerization

PROJECT GROUP
GROUP G55

PROJECT GUIDE
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Problem Statement

Legacy healthcare scheduling mechanisms rely on manual phone calls and paper registers, triggering critical operational bottlenecks:

- ✓ **Excessive Wait Times:** Unmanaged patient walk-ins cause heavy physical hospital congestion and delays.
- ✓ **Slot Collisions & Overbooking:** Absence of real-time synchronized doctor calendars across departments.
- ✓ **Diagnostic Inefficiencies:** Patients arrive unprepared for specialized medical packages (e.g., fasting missed).
- ✓ **Data Fragility:** Unstructured record storage prone to corruption, lacking ACID compliance and cloud resilience.

Project Objectives

- ✓ **Unified Patient Portal:** Interactive UI to explore doctor credentials, fees, and instant slot reservations.
- ✓ **Diagnostic Package Workflow:** Automated medical test scheduling with mandatory preparation instructions.
- ✓ **Executive Admin Dashboard:** Real-time doctor schedule control, appointment statuses, and analytics.
- ✓ **100% PostgreSQL ACID Storage:** Strict relational models (Sequelize ORM) with relational integrity.
- ✓ **Cloud-Native EC2 Deployment:** Multi-container Docker Compose with Nginx proxy on AWS Free Tier.

Technologies Used

- AWS EC2 Cloud
- Docker Engine
- Docker Compose
- PostgreSQL 15
- Sequelize ORM
- Nginx Proxy
- React 19
- Vite 7
- Node.js 20
- Express API
- Tailwind CSS
- Terraform (IaC)

Proposed Solution (MediCare SAAS)

A decoupled, cloud-orchestrated hospital scheduling ecosystem providing dedicated web interfaces for patients and hospital administrators backed by a high-throughput relational backend:

Client Tier	Nginx Gateway	Express API	PostgreSQL 15
Patient App (Port 80) Admin Portal (8080)	Reverse Proxy Dual-Port Routing	Node.js Engine Sequelize ORM	Relational DB Persistent Volumes

< 300ms	100% ACID	$0.00 / Mo
API LATENCY	DATA INTEGRITY	AWS FREE TIER

Architectural Highlights

- ✓ **Dual-Portal Micro-Routing:** Unified Nginx ingress routing public traffic on port 80 and hospital admin controls on port 8080.
- ✓ **Automated Swap Virtualization:** Configured 2GB swap space on AWS EC2 (t2.micro) ensuring smooth Vite asset builds.
- ✓ **Dynamic API Fallback:** Environment-aware API endpoint auto-detection eliminating cross-origin CORS faults.
- ✓ **Relational Data Models:** Strongly-typed tables for Doctors, Services, and Appointments with foreign keys.

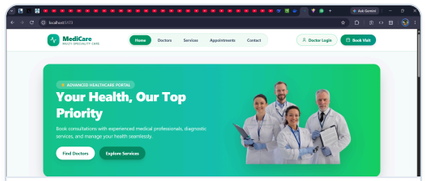
Source Code & Git Version Control

OFFICIAL GITHUB REPOSITORY
`https://github.com/VedantDhalkari/MediCare_SAAS.git`
Public • Branch: main

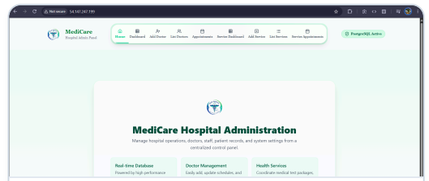
Conclusion

The *Cloud-Based Hospital Appointment System* replaces physical queues with a containerized platform. Built with PostgreSQL and AWS, the system proves that enterprise-grade medical scheduling can be deployed securely and reliably at zero hosting cost.

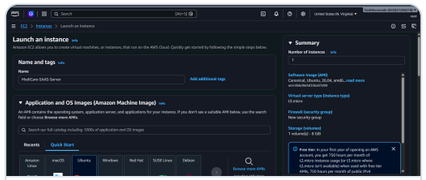
Live System Screenshots & Cloud Proof



Patient Portal (Port 80) — React 19



Admin Dashboard (Port 8080) — PostgreSQL Active



AWS EC2 Instance Launch — Free Tier


Live REST API (/api/doctors) — PostgreSQL 15

Results & Key Outcomes

- ✓ **Live Specialist Directory:** 8 pre-seeded medical specialties with bios, fees, and booking slots.
- ✓ **Diagnostic Booking:** Full-body checkups and cardiac packages with preparation instructions.
- ✓ **Administrative Analytics:** Real-time dashboard showing appointment metrics (Completed, Confirmed, Pending).
- ✓ **Automated 1-Command Deployment:** Shell-automated deployment script (`deploy.sh`) and instant git sync (`update.sh`).